A BASIS OF FREE GENERALIZED GERSTENHABER ALGEBRAS

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Abstract. Generalized Gerstenhaber algebras are the odd generalized Poisson superalgebras of Cantarini and Kac; Gerstenhaber algebras (odd Poisson superalgebras) form the special case in which the map $\mathfrak{D}(a) = \{1, a\}$ vanishes. We construct a linear basis of the free unital generalized Gerstenhaber algebra generated by a linearly ordered set of even and odd generators, following the method used earlier for free generalized Poisson superalgebras. As an application we show that the multilinear part of the free generalized Gerstenhaber algebra on n generators has dimension $n \cdot n!$.

1. Introduction

The notion of a Poisson bracket has its origin in the works of S. D. Poisson on celestial mechanics at the beginning of the XIX century. Since then Poisson algebras have appeared in several areas of mathematics, such as Poisson manifolds [23], algebraic geometry [4, 12], noncommutative geometry [29], operads, quantization theory [21], quantum groups, classical and quantum mechanics. The study of Poisson algebras has also led to a number of related algebraic structures: generic Poisson algebras, algebras of Jordan brackets and generalized Poisson algebras, Novikov–Poisson algebras, Malcev–Poisson–Jordan algebras, transposed Poisson algebras, n-ary Poisson algebras, Gerstenhaber algebras [22], and others.

A Gerstenhaber algebra (also called an odd Poisson superalgebra, or an anti-Poisson algebra) is a supercommutative associative superalgebra equipped with a bracket of degree -1 which is a Lie superalgebra bracket with respect to the reversed parity and acts on the associative product by derivations. Such brackets arise in physics, most notably in the Batalin–Vilkovisky formalism [2, 7, 3], and in geometry [33]. One of the most important purely algebraic sources of Gerstenhaber algebras is the Hochschild cohomology of an associative algebra [11]; for recent results on Gerstenhaber brackets on Hochschild cohomology see [32, 30, 31, 24]. Subalgebras of the Gerstenhaber algebra of differential operators are described in [6].

Generalized Poisson superalgebras appeared in the study of Jordan superalgebras. Kantor [15, 16] associated to a Poisson superalgebra a Jordan superalgebra (the Kantor double), and King and McCrimmon [20] described the brackets whose Kantor double is a Jordan superalgebra; the resulting superalgebras of Jordan brackets are, up to a change of the bracket, exactly the generalized Poisson superalgebras (see [8, 18]; a generalized version of the Kantor double was studied in [17]). In a generalized Poisson superalgebra the Leibniz rule holds only up to a correction term involving the linear map $\mathfrak{D}(a) = \{1, a\}$. Simple linearly compact generalized Poisson superalgebras were classified by Cantarini and Kac [8], who later introduced odd generalized Poisson superalgebras and classified the simple linearly compact ones [9]. Odd generalized Poisson superalgebras with $\mathfrak{D} = 0$ are precisely Gerstenhaber algebras; accordingly, we call odd generalized Poisson superalgebras *generalized Gerstenhaber algebras*.

Bases of free algebras were constructed for free Lie algebras [13, 27, 10, 25], free Lie superalgebras [28, 1, 5], free Poisson superalgebras [26], free generic Poisson algebras [19], free generalized Poisson superalgebras and free superalgebras of Jordan brackets [18], and so on. The aim of the

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present note is to construct a basis of the free unital generalized Gerstenhaber algebra generated by a linearly ordered set of even and odd generators (Theorem 12) and to compute the dimension of its multilinear part (Theorem 14). We follow the method of Shirshov [27] in the form used in [18] for free generalized Poisson superalgebras: the candidate basis U consists of supercommutative monomials in the elements of a basis of the free odd Lie superalgebra, and the linear independence of U is proved by constructing a generalized Gerstenhaber algebra structure on the linear span of U. There is, however, a new phenomenon in the odd case: the element $\{1,1\}$, which vanishes automatically in every generalized Poisson superalgebra, is a nonzero basis element of the free generalized Gerstenhaber algebra, so that the map $\mathfrak{D}$ is no longer a derivation. To handle this we define the bracket on the linear span of U by an explicit closed formula (Section 4), for which skew-symmetry and the Leibniz rule are verified directly, and we derive the Jacobi identity from a general formula for the Jacobiator (Lemma 9), which may be of independent interest.

2. Definitions and notation

Throughout the paper $\mathbb{F}$ is a field of characteristic zero. All vector spaces and algebras are $\mathbb{Z}_2$-graded, and the elements appearing in formulas involving parities are assumed to be homogeneous. For a homogeneous element a we write $|a| \in \mathbb{Z}_2$ for its parity and

$$\|a\| = |a| + 1 \in \mathbb{Z}_2$$

for its parity with respect to the reversed $\mathbb{Z}_2$-grading.

Definition 1. A *generalized Gerstenhaber algebra* is a $\mathbb{Z}_2$-graded vector space $V = V_0 \oplus V_1$ with an element $1 \in V_0$ and two bilinear operations $(a,b) \mapsto ab$ and $(a,b) \mapsto \{a,b\}$ such that for all homogeneous $a,b,c \in V$

$$|ab| = |a| + |b|, \qquad |\{a,b\}| = |a| + |b| + 1; \tag{1}$$

$$(ab)c = a(bc), \qquad ab = (-1)^{|a||b|}ba, \qquad 1a = a; \tag{2}$$

$$\{a,b\} = -(-1)^{\|a\|\|b\|}\{b,a\}; \tag{3}$$

$$\{a,\{b,c\}\} = \{\{a,b\},c\} + (-1)^{\|a\|\|b\|}\{b,\{a,c\}\}; \tag{4}$$

$$\{a,bc\} = \{a,b\}c + (-1)^{\|a\|\|b\|}b\{a,c\} + (-1)^{\|a\|}\mathfrak{D}(a)bc, \qquad \text{where } \mathfrak{D}(a) = \{1,a\}. \tag{5}$$

A generalized Gerstenhaber algebra with $\mathfrak{D} = 0$ is called a *Gerstenhaber algebra*.

Thus the product is even, associative and supercommutative with unit 1, the bracket is odd, and (3), (4) are the odd skew-symmetry and the odd Jacobi identity, while (5) is a generalized odd Leibniz rule. Since $\|a\|\|b\| = (|a|-1)(|b|-1)$ in $\mathbb{Z}_2$, Definition 1 is exactly the definition of an odd generalized Poisson superalgebra given in [9, (0.14)], and Gerstenhaber algebras are exactly odd Poisson superalgebras. The basic examples of generalized Gerstenhaber algebras with $\mathfrak{D} \neq 0$ are the superalgebras $PO(n, n+1)$ of [9].

Remark 2 (reversed parity). Let ΠV denote the space V with the reversed $\mathbb{Z}_2$-grading, so that the parity of a in ΠV is $\|a\|$. Identities (1), (3) and (4) say precisely that $(\Pi V, \{\,,\,\})$ is a Lie superalgebra. We call a superalgebra with an odd bilinear operation satisfying (3) and (4) an *odd Lie superalgebra*. Reversing the parity establishes a bijection between odd Lie superalgebras and Lie superalgebras (cf. [14] and [9, Remark 1.3]); in particular, the free odd Lie superalgebra generated by a set X of even and odd elements is the free Lie superalgebra generated by the same set with reversed parities.

Remark 3 (the map $\mathfrak{D}$). (a) Taking $a = 1$ in (5) we obtain

$$\mathfrak{D}(bc) = \mathfrak{D}(b)c + (-1)^{|b|}b\mathfrak{D}(c) - \mathfrak{D}(1)bc. \tag{6}$$

Hence $\mathfrak{D}$ is an odd derivation of $(V,\cdot)$ if and only if $\mathfrak{D}(1)=\{1,1\}=0$; in general, $\mathfrak{D}-\mathfrak{D}(1)\,\mathrm{id}$ is an odd derivation. In a generalized Poisson superalgebra (where the bracket is even) the element $\{1,1\}$ vanishes by skew-symmetry, but (3) imposes no condition on $\{1,1\}$ because $\|1\|=1$. We shall see (Corollary 13) that $\{1,1\}\neq 0$ in the free generalized Gerstenhaber algebra.

(b) Using (2) and $|\mathfrak{D}(a)|=\|a\|$, one checks that (5) is equivalent to the following statement: for every $a\in V$ the map

$$d_a\colon b\mapsto\{a,b\}+(-1)^{\|a\|}\mathfrak{D}(a)b$$

is a superderivation of parity $\|a\|$ of the superalgebra $(V,\cdot)$. In particular $d_a(1)=0$, i.e. $\{a,1\}=-(-1)^{\|a\|}\mathfrak{D}(a)$, in accordance with (3).

The following example shows that the identities of Definition 1 indeed do not force $\{1,1\}=0$.

Example 1. Let $V=\mathbb{F}1\oplus\mathbb{F}\varepsilon$ with $|1|=0$, $|\varepsilon|=1$, the product given by $1\cdot 1=1$, $1\varepsilon=\varepsilon 1=\varepsilon$, $\varepsilon^2=0$, and the bracket given by

$$\{1,1\}=\varepsilon,\qquad \{1,\varepsilon\}=\{\varepsilon,1\}=\{\varepsilon,\varepsilon\}=0.$$

Then V is a generalized Gerstenhaber algebra with $\mathfrak{D}(1)=\varepsilon\neq 0$ and $\mathfrak{D}(\varepsilon)=0$. Indeed, (1) and (2) are clear (note that $\varepsilon^2=0$ is forced by supercommutativity). The only nonzero bracket of basis elements is $\{1,1\}$, and (3) for $a=b=1$ reads $\{1,1\}=\{1,1\}$ because $\|1\|=1$. All brackets lie in $\mathbb{F}\varepsilon$ and $\{\varepsilon,V\}=\{V,\varepsilon\}=0$, so both sides of (4) vanish. It suffices to check (5) on basis elements. If $a=\varepsilon$, all terms vanish. If $a=1$ and $\varepsilon\in\{b,c\}$, every term contains either a bracket with ε or the product ε^2, hence vanishes. Finally, for $a=b=c=1$ the identity reads $\varepsilon=\varepsilon+\varepsilon-\varepsilon$. In this algebra $\mathfrak{D}(1\cdot 1)=\varepsilon\neq 2\varepsilon=\mathfrak{D}(1)1+1\mathfrak{D}(1)$, so $\mathfrak{D}$ is not a derivation, in accordance with (6).

3. The free generalized Gerstenhaber algebra and the set U

Let $X=\{1,x_1,x_2,\dots\}$ be a finite or countable set, linearly ordered by $1<x_1<x_2<\cdots$, in which 1 is even and each x_i has a prescribed parity $|x_i|\in\mathbb{Z}_2$. Let D_X be the free algebra with two bilinear operations $\cdot$ and $\{\,,\}$ generated by X; its basis consists of the nonassociative words in the alphabet X built with these two operations. The words are homogeneous, with $|1|=0$, $|uv|=|u|+|v|$ and $|\{u,v\}|=|u|+|v|+1$, and the *degree* of a word is the number of letters in it. Let I be the ideal of D_X generated by the elements

$$ab-(-1)^{|a||b|}ba,\qquad (ab)c-a(bc),\qquad 1a-a,\qquad \{a,b\}+(-1)^{\|a\|\|b\|}\{b,a\},$$
$$\{a,\{b,c\}\}-\{\{a,b\},c\}-(-1)^{\|a\|\|b\|}\{b,\{a,c\}\},$$
$$\{a,bc\}-\{a,b\}c-(-1)^{\|a\||b|}b\{a,c\}-(-1)^{\|a\|}\{1,a\}bc,$$

where a,b,c run over the homogeneous elements of D_X. Clearly, $\mathfrak{G}=D_X/I$ is the free unital generalized Gerstenhaber algebra generated by X: every parity preserving map from $\{x_1,x_2,\dots\}$ to a generalized Gerstenhaber algebra V extends uniquely to a homomorphism $\mathfrak{G}\to V$ sending 1 to 1.

Good words. Consider the nonassociative words in the alphabet X with respect to the single operation $\{\,,\}$. We define the set of *good words* together with a linear order on it by induction on the degree. The good words of degree 1 are the elements of X, ordered as above. Suppose that the good words of degree less than n are defined and linearly ordered. A word w of degree n is good whenever

(1) $w=\{u,v\}$, where u and v are good words with $u>v$;
(2) if $u=\{u_1,u_2\}$, then $u_2\le v$.

Every good word of degree n is greater than every good word of smaller degree; two good words $w=\{u_1,u_2\}$ and $w'=\{u_1',u_2'\}$ of degree n are compared lexicographically: $w>w'$ if $u_1>u_1'$, or $u_1=u_1'$ and $u_2>u_2'$. In particular, if $w=\{u,v\}$ is a good word, then $w>u>v$. Thus the good

words are the basic commutators of M. Hall [13]; with respect to the opposite order they form a Hall set in the sense of [25, Section 4.1].

The set M. Put

$$M = \{\, u,\ \{v,v\} \ \mid\ u,v \text{ are good words and } |v| = 0 \,\}.$$

We extend the order to M by the same rule: elements of larger degree are greater, and elements of the same degree are compared lexicographically by their pairs of components, a square $\{v,v\}$ being compared as the pair (v,v). This is a linear order on M (a square $\{v,v\}$ is never a good word, since the components $u_1 > u_2$ of a good word are distinct), and 1 is the least element of M. We enumerate the elements of M as $e_1 < e_2 < \cdots$, so that $e_1 = 1$.

Let $\mathfrak{L}_X$ be the free algebra with one bilinear operation $\{\,,\}$ generated by X (with the parities of words defined as above) and let J be the ideal of $\mathfrak{L}_X$ generated by the elements $\{a,b\} + (-1)^{\|a\|\|b\|}\{b,a\}$ and $\{a,\{b,c\}\} - \{\{a,b\},c\} - (-1)^{\|a\|\|b\|}\{b,\{a,c\}\}$, $a,b,c \in \mathfrak{L}_X$ homogeneous. The quotient $\mathfrak{L} = \mathfrak{L}_X/J$ is the free odd Lie superalgebra generated by X.

Lemma 4. *The set M is a basis of $\mathfrak{L}$.*

Proof. By Remark 2, $\mathfrak{L}$ is the free Lie superalgebra generated by the set X in which the parity of $x \in X$ is $\|x\|$. By [28, Theorem 1], the free Lie superalgebra over a field of characteristic zero generated by a linearly ordered set has a basis consisting of the basic commutators defined above and the squares $\{v,v\}$ of the odd ones; indeed, our good words are exactly the correct words of [28] for the order chosen above (cf. [18, Section 3]). For the Lyndon–Shirshov words the same result is proved in [1, Chapter 2] and [5]. An element v of $\mathfrak{L}$ is odd with respect to the reversed parity if and only if $|v| = 0$; hence this basis is exactly M. □

The set U. Let

$$U = \left\{\, e_{i_1}^{k_1} e_{i_2}^{k_2} \cdots e_{i_r}^{k_r} \ \mid\ r \geq 0,\ 1 < i_1 < i_2 < \cdots < i_r,\ k_j \geq 1, \text{ and } k_j = 1 \text{ if } |e_{i_j}| = 1 \,\right\}$$

be the set of supercommutative monomials in the elements of $M \setminus \{1\}$ (for $r = 0$ the empty product is 1). For an element $u = e_{i_1}^{k_1} \cdots e_{i_r}^{k_r}$ of U we call $\deg_e(u) = \sum_j k_j$ its *e-degree*; the elements of M are the elements of U of e-degree 1 together with 1. We regard the elements of M and U as elements of $\mathfrak{G}$ (evaluating the words in $\mathfrak{G}$). Since the product of $\mathfrak{G}$ is supercommutative and $ee = 0$ for odd e, the product of two elements of U is, up to sign, again an element of U or zero.

The main goal of the paper is to prove that U is a basis of $\mathfrak{G}$. We begin with the easier half.

Lemma 5. *The set U spans $\mathfrak{G}$.*

Proof. Since $(\mathfrak{G}, \{\,,\})$ satisfies (3) and (4), the identity map on X extends to a homomorphism $\psi\colon \mathfrak{L} \to \mathfrak{G}$ of odd Lie superalgebras; it sends each element of M to the same element of $\mathfrak{G}$. By Lemma 4, $\psi(\mathfrak{L})$ is spanned by $M \subset U$.

We first show that $\{u,v\} \in \operatorname{span} U$ for all $u,v \in U$, by induction on $N = \deg_e(u) + \deg_e(v)$, where we put $\deg_e(1) = 0$. If $u,v \in M$, then $\{u,v\} = \psi(\{u,v\}) \in \psi(\mathfrak{L}) \subset \operatorname{span} U$. Let $\deg_e(v) \geq 2$, so that $v = v_1 v_2$ with $v_1, v_2 \in U \setminus \{1\}$ and $\deg_e(v_1), \deg_e(v_2) < \deg_e(v)$. By (5),

$$\{u, v_1 v_2\} = \{u, v_1\} v_2 + (-1)^{\|u\|\|v_1\|} v_1 \{u, v_2\} + (-1)^{\|u\|} \{1, u\} v_1 v_2.$$

For each of the pairs (u, v_1), (u, v_2) and $(1, u)$ the sum of the e-degrees is less than N (for the last pair it equals $\deg_e(u) < \deg_e(u) + \deg_e(v)$), so $\{u, v_1\}$, $\{u, v_2\}$ and $\{1, u\}$ lie in $\operatorname{span} U$ by the induction hypothesis. Since $\operatorname{span} U$ is closed under the product, $\{u,v\} \in \operatorname{span} U$. Finally, if $\deg_e(v) \leq 1$ and $\deg_e(u) \geq 2$, then $\{u,v\} = -(-1)^{\|u\|\|v\|}\{v,u\}$ by (3), and $\{v,u\} \in \operatorname{span} U$ by the previous case, applied to the pair (v,u) with the same value of N.

Thus $\operatorname{span} U$ is closed under the bracket and, since U is closed under the product up to sign, also under the product. As $X \subset M \subset U$, the subspace $\operatorname{span} U$ is a subalgebra of $\mathfrak{G}$ containing the generators, hence $\operatorname{span} U = \mathfrak{G}$.

Note that the number of letters of a word is not a suitable induction parameter here, because it is not preserved by the defining relations of $\mathfrak{G}$: the relation $1a = a$ decreases it, while the last term of (5) increases it. For example, the expansion of the word $\{x_1, x_2x_3\}$ with four letters contains the term $(-1)^{\|x_1\|}\{1, x_1\}x_2x_3$, a nonzero multiple of the element $\{x_1, 1\}x_2x_3 \in U$ with five letters. □

4. The model algebra

In this section we construct a generalized Gerstenhaber algebra structure on the free supercommutative superalgebra generated by $M \setminus \{1\}$.

Let $\mathfrak{L}' = \operatorname{span}(M \setminus \{1\}) \subset \mathfrak{L}$, and let $\mathcal{S}$ be the free supercommutative superalgebra generated by the superspace $\mathfrak{L}'$, i.e. the tensor product of the symmetric algebra of $\mathfrak{L}'_0$ and the exterior algebra of $\mathfrak{L}'_1$. The monomials in the elements of the basis $M \setminus \{1\}$ of $\mathfrak{L}'$, that is, the elements of the set U, form a basis of $\mathcal{S}$. Put $\widehat{\mathfrak{L}} = \mathbb{F}1 \oplus \mathfrak{L}' \subset \mathcal{S}$. The linear map $\mathfrak{L} \to \widehat{\mathfrak{L}}$ which is the identity on $\mathfrak{L}'$ and sends the basis element $1 \in M$ to the unit $1 \in \mathcal{S}$ is an isomorphism of superspaces (by Lemma 4); we use it to transfer the bracket of $\mathfrak{L}$ to $\widehat{\mathfrak{L}}$. Thus $\widehat{\mathfrak{L}}$ is an odd Lie superalgebra with basis M, in which 1 is the least basis element, and $\widehat{\mathfrak{L}}$ generates $\mathcal{S}$ as a unital algebra. For $u \in \widehat{\mathfrak{L}}$ we put $\mathfrak{D}(u) = \{1, u\} \in \widehat{\mathfrak{L}}$; note that $\mathfrak{D}(1) = \{1, 1\}$ is an odd element of $M \setminus \{1\}$.

Notation. For a product $a = a_1a_2 \cdots a_m$ of homogeneous elements of $\mathcal{S}$ and $1 \le i \le m$ we write

$$a_{<i} = a_1 \cdots a_{i-1}, \qquad a_{>i} = a_{i+1} \cdots a_m, \qquad a_{\hat{\imath}} = a_{<i}a_{>i}$$

(empty products are equal to 1). By (2), $a_{<i}a_i = (-1)^{|a_i||a_{<i}|}a_ia_{<i}$ and $a = (-1)^{|a_i||a_{<i}|}a_ia_{\hat{\imath}} = (-1)^{|a_i||a_{>i}|}a_{\hat{\imath}}a_i$.

The bracket. For $a = a_1 \cdots a_m$ and $b = b_1 \cdots b_n$ with $m, n \ge 1$ and homogeneous $a_i, b_j \in \widehat{\mathfrak{L}}$ we put

$$\mathfrak{D}(a) = \sum_{i=1}^{m} (-1)^{|a_{<i}|}\, a_{<i}\{1, a_i\}a_{>i} - (m-1)\{1, 1\}\, a \tag{7}$$

and

$$\{a, b\}^* = \sum_{i=1}^{m}\sum_{j=1}^{n} (-1)^{|a_i||a_{>i}|+|b_j||b_{<j}|}\, a_{\hat{\imath}}\{a_i, b_j\}b_{\hat{\jmath}} \\ - (m-1)\, a\mathfrak{D}(b) + (n-1)(-1)^{\|a\|}\mathfrak{D}(a)b - (m-1)(n-1)\, a\mathfrak{D}(1)b. \tag{8}$$

Here $\{a_i, b_j\}$ is the bracket of $\widehat{\mathfrak{L}}$, so that $\{a, b\}^*$ is defined in terms of the bracket of $\mathfrak{L}$ and the product of $\mathcal{S}$. For $m = 1$ and $a_1 = 1$ formula (7) gives $\mathfrak{D}(1) = \{1, 1\}$, in agreement with the earlier notation, and formula (8) with $m = 1$, $a_1 = 1$ gives $\{1, b\}^* = \mathfrak{D}(b)$ (see Lemma 6). The first sum in (8) is the bracket obtained by moving a_i to the right end of a and b_j to the left end of b and bracketing the two adjacent factors; the remaining terms are corrections involving $\mathfrak{D}$.

Lemma 6. *The right-hand sides of (7) and (8) do not depend on the choice of the factorizations $a = a_1 \cdots a_m$ and $b = b_1 \cdots b_n$ with $a_i, b_j \in \widehat{\mathfrak{L}}$. Consequently, (7) defines a linear map $\mathfrak{D} \colon \mathcal{S} \to \mathcal{S}$ and (8) defines a bilinear map $\{\,,\}^* \colon \mathcal{S} \times \mathcal{S} \to \mathcal{S}$. Moreover,*

(i) *$\{u, v\}^* = \{u, v\}$ and $\mathfrak{D}(u) = \{1, u\}$ for $u, v \in \widehat{\mathfrak{L}}$;*
(ii) *$\{1, a\}^* = \mathfrak{D}(a)$ for all $a \in \mathcal{S}$;*
(iii) *$\mathfrak{D}(bc) = \mathfrak{D}(b)c + (-1)^{|b|}b\mathfrak{D}(c) - \mathfrak{D}(1)bc$ for all $b, c \in \mathcal{S}$;*
(iv) *$|\{a, b\}^*| = |a| + |b| + 1$ and $|\mathfrak{D}(a)| = \|a\|$ for all homogeneous $a, b \in \mathcal{S}$.*

Proof. Fix m and n. The right-hand side of (7) is a multilinear function $R_m(a_1, \ldots, a_m)$ of $a_1, \ldots, a_m \in \widehat{\mathfrak{L}}$, and the right-hand side of (8) is a multilinear function $R_{m,n}(a_1, \ldots, a_m; b_1, \ldots, b_n)$. We claim that these functions are

(a) supersymmetric: interchanging two neighbouring arguments a_k, a_{k+1} (or b_k, b_{k+1}) multiplies the value by $(-1)^{|a_k||a_{k+1}|}$ (respectively, by $(-1)^{|b_k||b_{k+1}|}$);
(b) compatible with removing units: $R_m(a_1,\dots,a_{m-1},1) = R_{m-1}(a_1,\dots,a_{m-1})$, $R_{m,n}(a_1,\dots,a_{m-1},1;b) = R_{m-1,n}(a_1,\dots,a_{m-1};b)$ for $m \geq 2$, and similarly in the second group of arguments.

Let us verify (a) for $R_{m,n}$ and the arguments a_k, a_{k+1}; the other cases are analogous or simpler. The products a, $a\mathfrak{D}(b)$ and $a\mathfrak{D}(1)b$ are multiplied by $(-1)^{|a_k||a_{k+1}|}$, and so is $\mathfrak{D}(a)b$ once (a) is known for R_m. In the double sum, the terms with $i \neq k, k+1$ are multiplied by $(-1)^{|a_k||a_{k+1}|}$, since $a_{\hat{\imath}}$ is and the exponent $|a_i||a_{>i}|$ is unchanged. The term with $i = k$ of the original sum is $(-1)^{|a_k|(|a_{k+1}|+|a_{>k+1}|)}a_{<k}a_{k+1}a_{>k+1}\{a_k,b_j\}b_{\hat{\jmath}}$, while the term with $i = k+1$ of the sum for the interchanged arguments is $(-1)^{|a_k||a_{>k+1}|}a_{<k}a_{k+1}a_{>k+1}\{a_k,b_j\}b_{\hat{\jmath}}$; these differ by the factor $(-1)^{|a_k||a_{k+1}|}$, and the same holds for the pair of terms with $i = k+1$ and $i = k$. For R_m the terms with $i \neq k, k+1$ and the last term of (7) are treated in the same way (the parity of $a_{<i}$ does not change). The term with $i = k$ of the original sum is $(-1)^{|a_{<k}|}a_{<k}\{1,a_k\}a_{k+1}a_{>k+1}$, while the term with $i = k+1$ of the sum for the interchanged arguments is

$$(-1)^{|a_{<k}|+|a_{k+1}|}a_{<k}a_{k+1}\{1,a_k\}a_{>k+1} = (-1)^{|a_{<k}|+|a_{k+1}|+|a_{k+1}|(|a_k|+1)}a_{<k}\{1,a_k\}a_{k+1}a_{>k+1},$$

since $|\{1,a_k\}| = |a_k|+1$; here the sign coming from the change of $(-1)^{|a_{<i}|}$ is compensated by the sign coming from moving $\{1,a_k\}$ past a_{k+1}, and the two terms differ by the factor $(-1)^{|a_k||a_{k+1}|}$. The pair of terms with $i = k+1$ and $i = k$ is treated in the same way.

To verify (b), let $a_m = 1$ and $a' = a_1\cdots a_{m-1}$. In (7) the term with $i = m$ equals $(-1)^{|a'|}a'\{1,1\} = \{1,1\}a'$ (recall that $\{1,1\}$ is odd), so $R_m(a_1,\dots,a_{m-1},1) = R_{m-1}(a_1,\dots,a_{m-1}) + \{1,1\}a' - \{1,1\}a'$. In (8) the terms with $i = m$ contribute $a'\sum_j (-1)^{|b_j||b_{<j}|}\{1,b_j\}b_{\hat{\jmath}}$, and since $(-1)^{|b_j||b_{<j}|}\{1,b_j\}b_{\hat{\jmath}} = (-1)^{|b_{<j}|}b_{<j}\{1,b_j\}b_{>j}$, this equals $a'\big(\mathfrak{D}(b) + (n-1)\mathfrak{D}(1)b\big)$ by (7); the terms with $i < m$ give the double sum for a'. The remaining terms of (8) for a and for a' differ by $-a'\mathfrak{D}(b) - (n-1)a'\mathfrak{D}(1)b$, which cancels the extra contribution of the double sum. The verification for $b_n = 1$ is similar: writing $b' = b_1\cdots b_{n-1}$ and using $\{a_i,1\} = -(-1)^{\|a_i\|}\{1,a_i\}$ (identity (3) in $\widehat{\mathfrak{L}}$), $a_{<i}\{1,a_i\}a_{>i} = (-1)^{\|a_i\||a_{>i}|}a_i\{1,a_i\}$ and (7), one finds that the terms with $j = n$ of the double sum contribute $(-1)^{|a|}\big(\mathfrak{D}(a) + (m-1)\mathfrak{D}(1)a\big)b'$, which cancels the difference $(-1)^{\|a\|}\mathfrak{D}(a)b' - (m-1)\,a\mathfrak{D}(1)b'$ between the remaining terms of (8) for b and for b'.

Now define $\mathfrak{D}$ on the basis U of $\mathcal{S}$ and $\{\,,\}^*$ on $U\times U$ by (7) and (8), using for $u = e_{i_1}^{k_1}\cdots e_{i_r}^{k_r} \in U$ the factorization into $\deg_e(u)$ factors from $M\setminus\{1\}$ taken in nondecreasing order, and for $u = 1$ the factorization with $m = 1$, $a_1 = 1$; extend by linearity. Then (7) and (8) hold for arbitrary factorizations: both sides are multilinear in the factors, so it suffices to consider factors in the basis M of $\widehat{\mathfrak{L}}$; by (a) the factors can be rearranged in nondecreasing order, both sides changing by the same sign; if some odd basis element occurs twice, the left-hand side is zero and so is the right-hand side by (a), because it changes sign under the interchange of two equal odd arguments; by (b) the factors equal to 1 can be removed, and what remains is the factorization used in the definition.

Statements (i), (ii) and (iv) follow directly from (7) and (8) with $m = n = 1$, respectively with $m = 1$, $a_1 = 1$ (using the identity $(-1)^{|b_j||b_{<j}|}\{1,b_j\}b_{\hat{\jmath}} = (-1)^{|b_{<j}|}b_{<j}\{1,b_j\}b_{>j}$ established above and $(-1)^{\|1\|} = -1$). For (iii), apply (7) to the factorization $bc = b_1\cdots b_n c_1\cdots c_p$: the terms corresponding to the factors of b give $(\mathfrak{D}(b) + (n-1)\mathfrak{D}(1)b)c$, the terms corresponding to the factors of c give $(-1)^{|b|}b(\mathfrak{D}(c) + (p-1)\mathfrak{D}(1)c) = (-1)^{|b|}b\mathfrak{D}(c) + (p-1)\mathfrak{D}(1)bc$, and the last term is $-(n+p-1)\mathfrak{D}(1)bc$. □

We now verify the identities (3) and (5) for $\{\,,\}^*$. In the proofs we write $\{a,b\}_0$ for the double sum in (8), so that

$$\{a,b\}^* = \{a,b\}_0 - (m-1)\,a\mathfrak{D}(b) + (n-1)(-1)^{\|a\|}\mathfrak{D}(a)b - (m-1)(n-1)\,a\mathfrak{D}(1)b. \tag{9}$$

Lemma 7. *For all $a, b \in \mathcal{S}$ we have $\{a,b\}^* = -(-1)^{\|a\|\|b\|}\{b,a\}^*$.*

Proof. Let $a = a_1 \cdots a_m$, $b = b_1 \cdots b_n$ with $a_i, b_j \in \widehat{\mathfrak{L}}$. Consider the term of $\{b,a\}_0$ corresponding to the pair (j,i):

$$(-1)^{|b_j||b_{>j}|+|a_i||a_{<i}|}\, b_{\hat{\jmath}}\{b_j, a_i\}a_{\hat{\imath}}.$$

By (3) in $\widehat{\mathfrak{L}}$, $\{b_j, a_i\} = -(-1)^{\|a_i\|\|b_j\|}\{a_i, b_j\}$, and by (2)

$$b_{\hat{\jmath}}\{a_i, b_j\}a_{\hat{\imath}} = (-1)^{|a_i|(|b_{\hat{\jmath}}|+\|a_i\|+|b_j|)+|b_j|(\|a_i\|+|b_j|)}\, a_{\hat{\imath}}\{a_i, b_j\}b_{\hat{\jmath}},$$

since $|\{a_i, b_j\}| = \|a_i\|+|b_j|$. Using $\|a\| = |a_i|+\|a_i\|$, $\|b\| = |b_{\hat{\jmath}}|+\|b_j\|$, $|b_{\hat{\jmath}}| = |b|+|b_j|$, $|a_{\hat{\imath}}| = |a|+|a_i|$ and $|a_i|+\|a_i\| = |b_j|+\|b_j\| = 1$, a direct comparison of the exponents shows that this term equals

$$-(-1)^{\|a\|\|b\|}\,(-1)^{|a_i||a_{>i}|+|b_j||b_{<j}|}\, a_{\hat{\imath}}\{a_i, b_j\}b_{\hat{\jmath}},$$

i.e. $-(-1)^{\|a\|\|b\|}$ times the (i,j)-term of $\{a,b\}_0$. Hence $\{b,a\}_0 = -(-1)^{\|a\|\|b\|}\{a,b\}_0$.

For the remaining terms of (9) we use $|\mathfrak{D}(a)| = \|a\|$, $|\mathfrak{D}(1)| = 1$ and (2):

$$b\mathfrak{D}(a) = (-1)^{|b|\|a\|}\mathfrak{D}(a)b, \qquad \mathfrak{D}(b)a = (-1)^{\|b\||a|}a\mathfrak{D}(b), \qquad b\mathfrak{D}(1)a = -(-1)^{\|a\|\|b\|}a\mathfrak{D}(1)b.$$

Therefore

$$\begin{aligned}&-(n-1)\,b\mathfrak{D}(a) + (m-1)(-1)^{\|b\|}\mathfrak{D}(b)a - (n-1)(m-1)\,b\mathfrak{D}(1)a\\ &\qquad = -(-1)^{\|a\|\|b\|}\Big(-(m-1)\,a\mathfrak{D}(b) + (n-1)(-1)^{\|a\|}\mathfrak{D}(a)b - (m-1)(n-1)\,a\mathfrak{D}(1)b\Big),\end{aligned}$$

because $(-1)^{\|b\|(|a|+1)} = (-1)^{\|a\|\|b\|}$ and $(-1)^{|b|\|a\|} = (-1)^{\|a\|\|b\|+\|a\|}$. Adding the two computations we obtain the lemma. □

Lemma 8. *For all $a, b, c \in \mathcal{S}$ we have*

$$\{a, bc\}^* = \{a,b\}^*c + (-1)^{\|a\||b|}b\{a,c\}^* + (-1)^{\|a\|}\mathfrak{D}(a)bc.$$

Proof. Let $a = a_1 \cdots a_m$, $b = b_1 \cdots b_n$, $c = c_1 \cdots c_p$ with $a_i, b_j, c_k \in \widehat{\mathfrak{L}}$, and apply (8) to the factorization $bc = b_1 \cdots b_n c_1 \cdots c_p$. The terms of $\{a, bc\}_0$ corresponding to the factors of b give $\{a,b\}_0 c$. A term corresponding to the factor c_k equals

$$(-1)^{|a_i||a_{>i}|+|c_k|(|b|+|c_{<k}|)}\, a_{\hat{\imath}}\{a_i, c_k\}\, b\, c_{\hat{k}} = (-1)^{\|a\||b|}\,(-1)^{|a_i||a_{>i}|+|c_k||c_{<k}|}\, b\, a_{\hat{\imath}}\{a_i, c_k\}c_{\hat{k}},$$

because $|a_{\hat{\imath}}\{a_i, c_k\}| = \|a\| + |c_k|$; hence these terms give $(-1)^{\|a\||b|}b\{a,c\}_0$. Thus

$$\{a, bc\}_0 = \{a,b\}_0 c + (-1)^{\|a\||b|}b\{a,c\}_0.$$

It remains to compare the terms involving $\mathfrak{D}$. By (9), on the left-hand side they are

$$-(m-1)\,a\mathfrak{D}(bc) + (n+p-1)(-1)^{\|a\|}\mathfrak{D}(a)bc - (m-1)(n+p-1)\,a\mathfrak{D}(1)bc,$$

and on the right-hand side they are

$$\begin{aligned}&\big(-(m-1)\,a\mathfrak{D}(b) + (n-1)(-1)^{\|a\|}\mathfrak{D}(a)b - (m-1)(n-1)\,a\mathfrak{D}(1)b\big)c\\ &+ (-1)^{\|a\||b|}b\big(-(m-1)\,a\mathfrak{D}(c) + (p-1)(-1)^{\|a\|}\mathfrak{D}(a)c - (m-1)(p-1)\,a\mathfrak{D}(1)c\big) + (-1)^{\|a\|}\mathfrak{D}(a)bc.\end{aligned}$$

Using (2) together with $|\mathfrak{D}(a)| = \|a\|$ and $|\mathfrak{D}(1)| = 1$ we have

$$(-1)^{\|a\||b|}ba\mathfrak{D}(c) = (-1)^{|b|}ab\mathfrak{D}(c), \quad (-1)^{\|a\||b|}b\mathfrak{D}(a)c = \mathfrak{D}(a)bc, \quad (-1)^{\|a\||b|}ba\mathfrak{D}(1)c = a\mathfrak{D}(1)bc,$$

so the right-hand side equals

$$-(m-1)\,a\big(\mathfrak{D}(b)c + (-1)^{|b|}b\mathfrak{D}(c)\big) + (n+p-1)(-1)^{\|a\|}\mathfrak{D}(a)bc - (m-1)(n+p-2)\,a\mathfrak{D}(1)bc,$$

which coincides with the left-hand side by Lemma 6(iii). □

To prove the Jacobi identity we use the following general formula for the Jacobiator.

Lemma 9. *Let $(V,\cdot,\{\,,\})$ be a superalgebra with unit satisfying (1), (2), (3) and (5), and let*

$$J(a,b,c)=\{a,\{b,c\}\}-\{\{a,b\},c\}-(-1)^{\|a\|\|b\|}\{b,\{a,c\}\}$$

for $a,b,c\in V$. Then

(i) $J(a,b,c)=-(-1)^{\|a\|\|b\|}J(b,a,c)=-(-1)^{\|b\|\|c\|}J(a,c,b)$;
(ii) $J(a,b,xy)=J(a,b,x)\,y+(-1)^{(|a|+|b|)|x|}\,x\,J(a,b,y)-J(a,b,1)\,xy$ *for all $a,b,x,y\in V$.*

Proof. (i) By (3), $\{\{b,a\},c\}=-(-1)^{\|a\|\|b\|}\{\{a,b\},c\}$, whence

$$-(-1)^{\|a\|\|b\|}J(b,a,c)=-(-1)^{\|a\|\|b\|}\{b,\{a,c\}\}-\{\{a,b\},c\}+\{a,\{b,c\}\}=J(a,b,c).$$

Similarly, $\{a,\{c,b\}\}=-(-1)^{\|b\|\|c\|}\{a,\{b,c\}\}$, $\{\{a,c\},b\}=-(-1)^{\|b\|(\|a\|+\|c\|)}\{b,\{a,c\}\}$ and $\{c,\{a,b\}\}=-(-1)^{\|c\|(\|a\|+\|b\|)}\{\{a,b\},c\}$, since $\|\{a,c\}\|=\|a\|+\|c\|$ and $\|\{a,b\}\|=\|a\|+\|b\|$; substituting these into $J(a,c,b)$ gives $J(a,c,b)=-(-1)^{\|b\|\|c\|}J(a,b,c)$.

(ii) We expand the three terms of $J(a,b,xy)$ by (5), using $|\{b,x\}|=\|b\|+|x|$, $|\mathfrak{D}(b)|=\|b\|$, $\|\{a,b\}\|=\|a\|+\|b\|$ and $(-1)^{(\|a\|+\|b\|)|x|}=(-1)^{(|a|+|b|)|x|}$:

$$\begin{aligned}
\{a,\{b,xy\}\}&=\{a,\{b,x\}\}y+(-1)^{\|a\|(|x|+\|b\|)}\{b,x\}\{a,y\}+(-1)^{\|a\|}\mathfrak{D}(a)\{b,x\}y\\
&\quad+(-1)^{\|b\||x|}\Big(\{a,x\}\{b,y\}+(-1)^{\|a\||x|}x\{a,\{b,y\}\}+(-1)^{\|a\|}\mathfrak{D}(a)x\{b,y\}\Big)\\
&\quad+(-1)^{\|b\|}\Big(\{a,\mathfrak{D}(b)\}xy+(-1)^{\|a\|\|b\|}\mathfrak{D}(b)\{a,xy\}+(-1)^{\|a\|}\mathfrak{D}(a)\mathfrak{D}(b)xy\Big),\\
\{\{a,b\},xy\}&=\{\{a,b\},x\}y+(-1)^{(|a|+|b|)|x|}x\{\{a,b\},y\}+(-1)^{\|a\|+\|b\|}\mathfrak{D}(\{a,b\})xy,\\
\{b,\{a,xy\}\}&=\{b,\{a,x\}\}y+(-1)^{\|b\|(|x|+\|a\|)}\{a,x\}\{b,y\}+(-1)^{\|b\|}\mathfrak{D}(b)\{a,x\}y\\
&\quad+(-1)^{\|a\||x|}\Big(\{b,x\}\{a,y\}+(-1)^{\|b\||x|}x\{b,\{a,y\}\}+(-1)^{\|b\|}\mathfrak{D}(b)x\{a,y\}\Big)\\
&\quad+(-1)^{\|a\|}\Big(\{b,\mathfrak{D}(a)\}xy+(-1)^{\|a\|\|b\|}\mathfrak{D}(a)\{b,xy\}+(-1)^{\|b\|}\mathfrak{D}(b)\mathfrak{D}(a)xy\Big),
\end{aligned}$$

where, in addition, $\{a,xy\}$ and $\{b,xy\}$ are to be expanded by (5). We substitute these expansions into $J(a,b,xy)=\{a,\{b,xy\}\}-\{\{a,b\},xy\}-(-1)^{\|a\|\|b\|}\{b,\{a,xy\}\}$. The terms $\{a,\{b,x\}\}y$, $-\{\{a,b\},x\}y$, $-(-1)^{\|a\|\|b\|}\{b,\{a,x\}\}y$ add up to $J(a,b,x)y$, and the terms containing $x\{a,\{b,y\}\}$, $x\{\{a,b\},y\}$, $x\{b,\{a,y\}\}$ add up to $(-1)^{(|a|+|b|)|x|}xJ(a,b,y)$. By (3) we have $\{b,1\}=-(-1)^{\|b\|}\mathfrak{D}(b)$, $\{a,1\}=-(-1)^{\|a\|}\mathfrak{D}(a)$ and $\{\{a,b\},1\}=-(-1)^{\|a\|+\|b\|}\mathfrak{D}(\{a,b\})$, so that

$$J(a,b,1)=-(-1)^{\|b\|}\{a,\mathfrak{D}(b)\}+(-1)^{\|a\|+\|b\|}\mathfrak{D}(\{a,b\})+(-1)^{\|a\|\|b\|+\|a\|}\{b,\mathfrak{D}(a)\};$$

hence the terms $(-1)^{\|b\|}\{a,\mathfrak{D}(b)\}xy$, $-(-1)^{\|a\|+\|b\|}\mathfrak{D}(\{a,b\})xy$, $-(-1)^{\|a\|\|b\|+\|a\|}\{b,\mathfrak{D}(a)\}xy$ add up to $-J(a,b,1)xy$. The remaining sixteen terms cancel in pairs: after the expansion of $\{a,xy\}$ and $\{b,xy\}$ they add up to

$$\begin{aligned}
&\big((-1)^{\|a\|(|x|+\|b\|)}-(-1)^{\|a\|\|b\|+\|a\||x|}\big)\{b,x\}\{a,y\}+\big((-1)^{\|b\||x|}-(-1)^{\|a\|\|b\|+\|b\|(|x|+\|a\|)}\big)\{a,x\}\{b,y\}\\
&+\big((-1)^{\|a\|}-(-1)^{\|a\|}\big)\mathfrak{D}(a)\{b,x\}y+\big((-1)^{\|b\||x|+\|a\|}-(-1)^{\|a\|+\|b\||x|}\big)\mathfrak{D}(a)x\{b,y\}\\
&+\big((-1)^{\|b\|+\|a\|\|b\|}-(-1)^{\|a\|\|b\|+\|b\|}\big)\mathfrak{D}(b)\{a,x\}y\\
&+\big((-1)^{\|b\|+\|a\|\|b\|+\|a\||x|}-(-1)^{\|a\|\|b\|+\|a\||x|+\|b\|}\big)\mathfrak{D}(b)x\{a,y\}\\
&+\big((-1)^{\|b\|+\|a\|}-(-1)^{\|a\|+\|b\|}\big)\mathfrak{D}(a)\mathfrak{D}(b)xy+\big((-1)^{\|b\|+\|a\|\|b\|+\|a\|}-(-1)^{\|a\|\|b\|+\|a\|+\|b\|}\big)\mathfrak{D}(b)\mathfrak{D}(a)xy,
\end{aligned}$$

and each of the eight coefficients is zero. □

Proposition 10. *The bracket $\{\,,\}^*$ satisfies the odd Jacobi identity (4) on $\mathcal{S}$.*

Proof. By Lemmas 6, 7 and 8, the superalgebra $(\mathcal{S}, \cdot, \{\,,\,\}^*)$ satisfies the hypotheses of Lemma 9; let J be its Jacobiator. We have to show that $J(a,b,c) = 0$ for all $a, b, c \in \mathcal{S}$.

First, for fixed $a, b \in \mathcal{S}$, if $J(a,b,u) = 0$ for all $u \in \widehat{\mathfrak{L}}$, then $J(a,b,c) = 0$ for all $c \in \mathcal{S}$. Indeed, $\mathcal{S}$ is spanned by the products $c = c_1 \cdots c_m$ of elements of $\widehat{\mathfrak{L}}$, and by Lemma 9(ii),

$$J(a,b,c_1\cdots c_m) = J(a,b,c_1\cdots c_{m-1})\,c_m + (-1)^{(|a|+|b|)|c_1\cdots c_{m-1}|}\,c_1\cdots c_{m-1}\,J(a,b,c_m) - J(a,b,1)\,c_1\cdots c_m,$$

so the claim follows by induction on m, since $1, c_m \in \widehat{\mathfrak{L}}$.

Now let $u, v, w \in \widehat{\mathfrak{L}}$. By Lemma 6(i), $J(u,v,w)$ is the Jacobiator of the odd Lie superalgebra $\widehat{\mathfrak{L}}$, hence $J(u,v,w) = 0$. By the previous paragraph, $J(u,v,c) = 0$ for all $c \in \mathcal{S}$. By Lemma 9(i), $J(u,c,v) = -(-1)^{\|c\|\|v\|}J(u,v,c) = 0$ for $u, v \in \widehat{\mathfrak{L}}$, $c \in \mathcal{S}$; applying the previous paragraph once more we get $J(u,b,c) = 0$ for all $u \in \widehat{\mathfrak{L}}$ and $b, c \in \mathcal{S}$. Finally, by Lemma 9(i), $J(a,b,u) = -(-1)^{\|b\|\|u\|}J(a,u,b) = (-1)^{\|b\|\|u\|+\|a\|\|u\|}J(u,a,b) = 0$ for $a, b \in \mathcal{S}$, $u \in \widehat{\mathfrak{L}}$, and the first paragraph gives $J(a,b,c) = 0$ for all $a, b, c \in \mathcal{S}$. □

Summarizing, we obtain

Theorem 11. *$(\mathcal{S}, \cdot, \{\,,\,\}^*)$ is a unital generalized Gerstenhaber algebra, whose bracket restricted to $\widehat{\mathfrak{L}}$ is the bracket of the free odd Lie superalgebra $\mathfrak{L}$ and whose map $\mathfrak{D}$ is given by (7). It is generated by X as a unital generalized Gerstenhaber algebra.*

Proof. The identities (1) and (2) hold in $\mathcal{S}$ by construction and by Lemma 6(iv); (3), (5) and (4) are Lemma 7, Lemma 8 (note that $\mathfrak{D}(a) = \{1, a\}^*$ by Lemma 6(ii)) and Proposition 10. The subalgebra of $\mathcal{S}$ generated by X contains $\widehat{\mathfrak{L}}$, because $\mathfrak{L}$ is generated by X as an odd Lie superalgebra and $\{\,,\,\}^*$ coincides with the bracket of $\mathfrak{L}$ on $\widehat{\mathfrak{L}}$; and $\widehat{\mathfrak{L}}$ generates $\mathcal{S}$ under the product. □

5. Main results

Theorem 12. *The set U is a basis of the free unital generalized Gerstenhaber algebra $\mathfrak{G}$ generated by X. Moreover, the homomorphism $\varphi\colon \mathfrak{G} \to \mathcal{S}$ which is the identity on X is an isomorphism of generalized Gerstenhaber algebras.*

Proof. By Theorem 11 and the universal property of $\mathfrak{G}$, the identity map on X extends to a homomorphism $\varphi\colon \mathfrak{G} \to \mathcal{S}$ of unital generalized Gerstenhaber algebras, which is surjective because $\mathcal{S}$ is generated by X. Every element $e \in M$ is a word in the generators with respect to the bracket, and φ is compatible with the brackets; hence $\varphi(e) = e$ for $e \in M$, and consequently $\varphi(u) = u$ for every $u \in U$. Since U is a basis of $\mathcal{S}$, the set $U \subset \mathfrak{G}$ is linearly independent, and by Lemma 5 it spans $\mathfrak{G}$. Thus U is a basis of $\mathfrak{G}$, and φ, being a surjection mapping a basis bijectively onto a basis, is an isomorphism. □

Corollary 13. *The element $\mathfrak{D}(1) = \{1,1\}$ is nonzero in $\mathfrak{G}$, and $\mathfrak{D}$ is not a derivation of $(\mathfrak{G}, \cdot)$. More precisely, the elements $\{1,1\}$, $\{1,1\}x_i$ and $\{1,x_i\}$, $i \geq 1$, are linearly independent in $\mathfrak{G}$.*

Proof. The words $\{1,1\}$ (a square of the even good word 1), $\{x_i, 1\}$ (a good word) and the monomials $\{1,1\}x_i$ belong to U; note that $\{1,x_i\} = -(-1)^{\|x_i\|}\{x_i,1\}$. The second assertion follows from Remark 3(a). The first assertion can also be seen directly, without Theorem 12: by the universal property of $\mathfrak{G}$ there is a homomorphism from $\mathfrak{G}$ to the algebra V of Example 1 sending 1 to 1 and every x_i to 0, and it maps $\{1,1\}$ to $\varepsilon \neq 0$. □

Following [18], for $n \geq 1$ consider the free unital generalized Gerstenhaber algebra $\mathfrak{G}$ generated by $X = \{1, x_1, \ldots, x_n\}$ (with arbitrary parities of $x_1, \ldots, x_n$) and denote by $\mathrm{GenGer}(n)$ the subspace of $\mathfrak{G}$ spanned by the elements of the basis U which are multilinear in $1, x_1, \ldots, x_n$, that is, in which each of the letters $1, x_1, \ldots, x_n$ occurs exactly once. Note that $\mathfrak{G}$ is graded by the multidegree in

$x_1, \dots, x_n$, but not by the number of occurrences of the letter 1: the relation $1a = a$ and the last term of (5) are not homogeneous in 1. Therefore $\mathrm{GenGer}(n)$ is defined as the span of a subset of the basis U (equivalently, via the isomorphism φ of Theorem 12, of monomials of $\mathcal{S}$), and not as a homogeneous component of $\mathfrak{G}$; it is a subspace of the component of $\mathfrak{G}$ of multidegree $(1, \dots, 1)$ in $x_1, \dots, x_n$.

Theorem 14. $\dim \mathrm{GenGer}(n) = n \cdot n!$.

Proof. An element of U is a product $e_{i_1} \cdots e_{i_r}$ of elements of $M \setminus \{1\}$, and it is multilinear in $1, x_1, \dots, x_n$ if and only if the factors are good words (squares $\{v, v\}$ contain every letter of v twice) whose sets of letters are pairwise disjoint and multilinear, with union X. Thus the multilinear elements of U are in bijection with the pairs consisting of a partition π of the set X into blocks none of which equals $\{1\}$ (the factor 1 is excluded in U) and a family of multilinear good words w_B, $B \in \pi$, where the set of letters of w_B is B.

The multilinear good words with a given set B of k letters form, by Lemma 4, a basis of the multilinear component of the free odd Lie superalgebra generated by B; since good words are defined purely combinatorially, their number does not depend on the parities of the letters, and it is equal to the dimension $(k-1)!$ of the multilinear component of the free Lie algebra of rank k (see [25]). Consequently,

$$\dim \mathrm{GenGer}(n) = \sum_{\pi} \prod_{B \in \pi} (|B| - 1)!,$$

where π runs over the partitions of X without the block $\{1\}$. For every finite set Y we have $\sum_\pi \prod_{B\in\pi}(|B|-1)! = |Y|!$, the sum being over all partitions of Y: a partition of Y together with a cyclic order on each block is the same as a permutation of Y (the blocks being its cycles). Subtracting the partitions of X containing the block $\{1\}$, which correspond to the partitions of $\{x_1, \dots, x_n\}$, we obtain $\dim \mathrm{GenGer}(n) = (n+1)! - n! = n \cdot n!$. □

Example 2. For $n = 1$ the only multilinear element of U is the good word $\{x_1, 1\}$, so $\dim \mathrm{GenGer}(1) = 1 = 1 \cdot 1!$. For $n = 2$ the multilinear good words with the set of letters $\{1, x_1, x_2\}$ are $\{\{x_1, 1\}, x_2\}$ and $\{\{x_2, 1\}, x_1\}$ (the word $\{\{x_2, x_1\}, 1\}$ is not good, since $x_1 > 1$), and the partitions $\{1, x_1\} \cup \{x_2\}$ and $\{1, x_2\} \cup \{x_1\}$ give the products $\{x_1, 1\}x_2$ and $\{x_2, 1\}x_1$. Hence $\mathrm{GenGer}(2)$ has the basis

$$\{x_1, 1\}x_2, \qquad \{x_2, 1\}x_1, \qquad \{\{x_1, 1\}, x_2\}, \qquad \{\{x_2, 1\}, x_1\},$$

and $\dim \mathrm{GenGer}(2) = 4 = 3! - 2! = 2 \cdot 2!$, in accordance with Theorem 14.

Remark 15. The same argument, applied to the free odd Lie superalgebra $\mathfrak{L}_0$ generated by $\{x_1, x_2, \dots\}$ with the bracket extended to $\mathbb{F}1 \oplus \mathfrak{L}_0$ by $\{1, \cdot\} = \{\cdot, 1\} = 0$, gives $\mathfrak{D} = 0$ in (7) and turns (8) into the unique extension of the bracket of $\mathfrak{L}_0$ to a biderivation of the free supercommutative superalgebra generated by $\mathfrak{L}_0$. Lemmas 6–9 and Proposition 10 remain valid verbatim, and the proof of Lemma 5 goes through with $\{1, u\} = 0$. This recovers the well-known description of the free unital Gerstenhaber algebra generated by $x_1, x_2, \dots$: its basis is formed by the elements of U in which the letter 1 does not occur.

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